

Challenges related to Ducts, Chimneys and Tail-end Alignment

Condition assessment (thickness measurement and weld point assessment) of ducts at inaccessible areas

Ducts of diameter ranging from few inches to ~5 meters spanning several kilometers across a typical manufacturing plant carry various gases (few of these are toxic). Ducts could be cylindrical, with or without external ribs, platform/flanges and, also having several bends. As part of periodic maintenance, the residual wall thickness of duct (range – 250mm to 500mm) is to be measured at critical locations (currently measured at 4 points - 12pm, 3pm, 6pm & 9pm positions) and is repaired/replaced based on the reduction in wall thickness/other defects. Also, these ducts are welded structures, so weld point thickness/weldment condition to be measured/assessed periodically.

Currently wall thickness measurement of the duct is carried out manually and involves skilled manpower working extensively based on requirement. The ducts could be laid overhead up to a height of 80 meters and the temperature of the duct could range from ambient to ~200 degC. The current practice involves a person working at height which poses a safety risk. Few cases, these measurements even not possible with rope access.

We are looking for a solution that can measure the duct wall thickness along the entire circumference (vertical as well as horizontal duct). Welded joints, if any, also need to be inspected. It will be an added advantage if the solution can be used for continuous measurement of the above-mentioned parameters.